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Estimating a War of Attrition: The Case of the US Movie Theater Industry

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1 Big picture

- **Question.** How much does strategic interaction delay firm exit in a declining industry, relative to a coordinated benchmark and to an industry-profit-maximizing benchmark?
- **Setting.** The paper studies the U.S. movie theater industry in the late 1940s and 1950s, when demand declined sharply because television diffusion reduced movie attendance.
- **Core challenge.** Exit timing is strategic: each theater prefers that rivals leave first. At the same time, exit values are private information, so theaters learn from survival and revise planned exit times as the market shrinks.
- **Main modeling contribution.** The paper extends the Fudenberg–Tirole war-of-attrition model from duopoly to oligopoly and derives a tractable symmetric perfect Bayesian equilibrium in continuous time.
- **Main empirical contribution.** It estimates a dynamic exit game using county-level markets, rich variation in TV penetration, and an auxiliary entry model to recover the base demand and the initial conditions of the exit game.
- **Main quantitative result.** Strategic interactions generate an average delay of exit of about **2.7 years**. But only a small part of this comes from strategic waiting due to asymmetric information: about **3.7%** of the delay is attributed to strategic behavior, while about **96.3%** is due to oligopolistic competition itself.
- **Main substantive lesson.** The cost of strategic interaction is largest in more concentrated markets and in markets where demand declines slowly. The observed sluggish consolidation of the theater industry is therefore not just a passive response to falling demand; it is shaped by strategic incentives.
- **Why this paper matters.** It is one of the clearest structural papers on exit timing in declining industries and provides a template for estimating wars of attrition in nonstationary oligopolies.

2 Data and measurement (key objects)

2.1 Observed theater-level panel

- The main data source is *The Film Daily Yearbook of Motion Pictures* for 1949–1952, 1954, and 1955, augmented with 1956 and 1957 volumes to clean spurious exits.
- The yearbooks list essentially every theater in the United States and provide theater name, location, number of seats when available, and type (indoor, drive-in, etc.).
- The paper defines as players in the exit game all theaters operating in 1949. The focus is on classic single-screen theaters rather than the new post-1949 entrants, which were often newer and systematically different facilities.
- Exit is not directly dated in the source. A theater is treated as having exited between year t and $t + 1$ if it is observed in year t and then absent, subject to a correction rule that checks whether it reappears in $t + 2$ or $t + 3$ with the same name and location.

- The final estimation sample contains **2,606 county markets** with at most 10 theaters in 1949, covering **9,768 theaters** initially.

2.2 Observed market-level state variables

- **Local market.** A market is a county. This choice allows the researcher to merge theater counts with county-level TV diffusion and demographic variables.
- **Demand shifter.** The key time-varying shifter is TV penetration, measured as the share of households owning at least one TV. Rates are observed in 1950, 1953, 1954, and 1955 and then interpolated/extrapolated using a Weibull CDF.
- **Demographics.** The paper uses 1950 county characteristics—population, median age, median family income, urban share, employment share, and land area—to capture the base demand for movie theaters.
- **Population change.** The growth rate of county population from 1950 to 1960 is used to shift the speed of demand decline.
- **New entry.** The number of post-1949 entrants in the market is also allowed to affect the speed with which demand for incumbent theaters declines.

2.3 Key measured objects and descriptive facts

- The nationwide number of indoor theaters fell from **17,367 in 1949** to **11,335 in 1959**.
- Average attendance per theater fell from about **261,991** admissions in 1949 to about **163,689** in 1955.
- In the estimation sample, there are many monopoly and duopoly markets: 17.3% of counties are monopolies, 20.0% are duopolies, 17.1% are triopolies, and almost 80% of counties have five theaters or fewer.
- TV diffusion is initially limited but becomes very heterogeneous across markets. In 1950, 87% of markets have TV penetration below 10%; by 1955, the 5th and 95th percentiles are roughly 14% and 86%.
- New entry after 1949 is limited: 61.8% of markets see no entry, 24.6% see one entrant, and only 13.6% see more than one entrant over the sample period.
- Exit rates are higher in markets with more competitors. Preliminary reduced-form evidence also shows that theaters' exits tend to be clustered in time.

2.4 Unobservables and timing

- Each theater i has a privately observed exit value θ_i , which captures both fixed operating costs and the value of the outside option.
- Each market m has unobserved base-demand heterogeneity α_m and unobserved heterogeneity in the speed of demand decline λ_m .

- Time is continuous. At every instant, theaters decide whether to remain or exit. Once a theater exits, it cannot reenter.
- When one theater exits, the surviving theaters observe that exit, update beliefs, and revise their planned exit times because the profit of staying rises when competition falls.

3 Models and empirical specifications (all equations + interpretation)

3.1 Continuous-time exit game

There are N theaters in a local market. Time starts at $t = 0$ and is continuous. If $n \in \{1, \dots, N\}$ theaters remain active, each surviving theater earns a common instantaneous profit $\Pi_n(t)$. Theater i has a privately known exit value θ_i , drawn i.i.d. from a distribution G on $[\underline{\theta}, \bar{\theta}]$, and discounts at rate r .

- **Economic meaning of θ_i .** A larger θ_i means theater i has a better outside option or a higher fixed cost of continuing.
- **Why private information matters.** Competitors do not know who is weak or strong; they infer this from who survives.

3.2 State space and strategy

The theater's state consists of its private exit value and a commonly observed component,

$$\omega^t = \{n, t, h^t\},$$

where n is the number of active theaters, t is current time, and h^t is the history of exits up to time t .

A strategy is a mapping from the state to a planned exit time,

$$T_i : S \times [\underline{\theta}, \bar{\theta}] \rightarrow \mathbb{R}_+.$$

In symmetric equilibrium, the player subscript is omitted and theaters with higher θ exit weakly earlier.

3.3 Assumptions on profits

The paper imposes four key restrictions on instantaneous profits:

- (i) $\Pi_n(t)$ decreases over time and converges to $\bar{\Pi}_n$,
- (ii) $\Pi_n(t) < \Pi_{n-1}(t)$ for $n \geq 2$,
- (iii) $\bar{\Pi}_n > \underline{\theta}$,
- (iv) $\Pi_1(0) < \bar{\theta}$.

- **Interpretation.** Demand declines over time, competition reduces profits, some theaters may survive forever in favorable states, but some theater wants to exit immediately at the start of the game.

3.4 Equilibrium concept

Let $V_i(\tau, T_{-i}, \omega, \theta)$ be the present discounted payoff to theater i if it chooses stopping time τ and the others follow strategy profile T_{-i} . Let $g(\theta_j | h^t)$ be player i 's posterior belief about rival j 's exit value given survival history. A symmetric *perfect Bayesian equilibrium* requires

$$V_i(T_i, T_{-i}, \omega, \theta) \geq V_i(\tau_i, T_{-i}, \omega, \theta)$$

for every theater and every feasible deviation, with beliefs updated by Bayes' rule whenever possible.

- **Interpretation.** The paper does not estimate a reduced-form hazard. It estimates an equilibrium stopping rule consistent with optimal behavior and belief updating.

3.5 Threshold policy and the marginal type

The equilibrium is characterized by a policy function $\Phi(t; \omega) = T^{-1}(t; \omega)$ giving the marginal exit value that is just indifferent between waiting and exiting at time t .

Before any exit occurs, the marginal type satisfies boundary conditions of the form

$$\Phi(0; \omega) = \Pi_{N-1}(0), \quad \lim_{t \rightarrow \infty} \Phi(t; \omega) = \bar{\Pi}_N.$$

After one theater exits, the game restarts with $n - 1$ players, a new state ω' , and a new threshold path with analogous boundary conditions.

- **Interpretation.** The equilibrium can be solved recursively: first solve the N -player phase; when one theater exits, move to the $(N - 1)$ -player subgame; continue until monopoly.

3.6 Marginal indifference condition

The economic content of the threshold rule comes from the indifference condition for the marginal type. For a theater just indifferent at time t , the expected benefit of waiting an instant equals the cost of waiting an instant. In reduced form,

$$(n - 1)\lambda(t; \omega) \left[V(t, \omega', \theta') - K \right] = \theta' - \Pi_n(t),$$

where $\lambda(t; \omega)$ is the conditional probability that one rival exits in the next instant, ω' is the continuation state after one rival exits, and K is the exit payoff term.

- **Left-hand side.** Expected gain from waiting because a rival may quit first.
- **Right-hand side.** Flow cost of waiting: the outside option forgone minus current operating profit.
- **Key insight.** The war of attrition lasts until the marginal type no longer finds it worth waiting for someone else to blink first.

3.7 Monopoly problem

When only one theater remains, there is no more strategic interaction. The last surviving theater solves a standard optimal stopping problem,

$$T(\omega', \theta) \in \arg \max_{t \in [t', \infty)} \left\{ \int_{t'}^t \Pi_1(s) e^{-r(s-t')} ds + \frac{\theta}{r} e^{-r(t-t')} \right\}.$$

- **Interpretation.** The last exit is not delayed by strategic interaction. This is why delays mainly concern earlier exits.

3.8 Profit specification for estimation

For market m , the paper parameterizes theater profits as

$$\Pi_n(t, m) = \pi_n(m) \cdot d(t, m),$$

where $\pi_n(m)$ is the market's base demand and $d(t, m)$ is a decay function capturing the time path of demand decline.

The base-demand component is specified as

$$\pi_n(m) = \frac{\alpha_m + \beta' X_m}{n^\delta}, \quad (1)$$

where X_m includes a constant, population, median age, median family income, urban share, employment share, and land area, while α_m is unobserved market heterogeneity.

The decay function is specified as

$$d(t, m) = \{1 - \lambda_1 TV_{tm}\}^{\exp(\lambda_m + \lambda_2 \Delta POP_m + \lambda_3 NEW_m)}, \quad (2)$$

with $0 \leq \lambda_1 \leq 1$, where TV_{tm} is TV penetration, ΔPOP_m is population growth from 1950 to 1960, NEW_m is the total number of entrants during the sample period, and λ_m is unobserved heterogeneity in the speed of decline.

- **Interpretation of δ .** It governs how much competition erodes profits.
- **Interpretation of the decay term.** TV diffusion is the main source of falling demand; population growth and new entry change the steepness of decline.

3.9 Distributional assumptions

Theater exit values follow a truncated normal distribution,

$$\theta_i \sim TN(\mu_\theta, \sigma_\theta^2; \ell_\theta, h_\theta),$$

and market heterogeneity terms are also normally distributed,

$$\alpha_m \sim N(\mu_\alpha, \sigma_\alpha^2), \quad \lambda_m \sim N(\mu_\lambda, \sigma_\lambda^2),$$

with possible correlation $\rho_{\alpha\lambda}$ between α_m and λ_m .

- **Interpretation.** Markets with high initial demand may also shrink faster or slower; $\rho_{\alpha\lambda}$ allows this correlation.

3.10 Auxiliary entry model

To recover the initial conditions of the 1949 exit game, the paper jointly models entry immediately before the war of attrition begins. Potential entrants decide simultaneously whether to enter.

If n theaters enter market m , each entrant earns

$$\pi_n^e(m) = \frac{\alpha_m + \beta' X_m}{n^{\delta_e}},$$

while a nonentrant receives its private outside value θ_i .

The optimal entry decision is

$$D_i^* = \mathbf{1} \left\{ (\alpha_m + \beta' X_m) \mathbb{E}[n_m^{-\delta_e}] \geq \theta_i \right\}.$$

Let P denote the symmetric entry probability. Then equilibrium entry beliefs solve the fixed-point equation

$$P = G\left((\alpha_m + \beta' X_m)K(P)\right), \quad (3)$$

where $K(P)$ is a decreasing function implied by the expected competition term.

- **Interpretation.** This entry stage helps identify the base-demand function and disciplines the joint distribution of market heterogeneity and incumbent exit values conditional on the number of initial theaters.

3.11 Indirect-inference estimator

Let γ denote the vector of structural parameters and let p denote the vector of empirical moments. For any candidate γ , the model is simulated to produce corresponding moments $p(\gamma)$. The estimator solves

$$\hat{\gamma} = \arg \min_{\gamma} (p - p(\gamma))' \Omega (p - p(\gamma)), \quad (4)$$

where Ω is a positive definite weighting matrix.

The asymptotic distribution is

$$\sqrt{M}(\hat{\gamma} - \gamma_0) \xrightarrow{d} N(0, W), \quad (5)$$

with M the number of markets.

- **Implementation.** The paper uses 28 moments to estimate 15 parameters. Moments come jointly from the entry model and the dynamic exit game.
- **Why indirect inference?** The full-solution approach exploits the unique symmetric equilibrium and avoids trying to estimate nonparametric conditional choice probabilities in a short panel.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- A simple regression of exit on market structure and TV diffusion is informative but not causal by itself.

- Markets with more theaters may simply be bigger or more attractive markets, so the number of competitors is endogenous.
- Unobserved market heterogeneity can affect both the initial number of theaters and the later speed of exit.
- Initial incumbents are a selected sample because only firms that chose to enter before 1949 appear in the exit game.

4.2 How the entry model identifies base demand and initial conditions

- Cross-market variation in the initial number of theaters helps identify the level of market profitability before the war of attrition begins.
- The auxiliary entry model uses exactly this variation to recover the demand shifters β and the entry-stage competition parameter δ_e .
- Because entry depends on private exit values, the entry model also helps recover the distribution of incumbent types conditional on the observed initial market structure.

4.3 How TV diffusion identifies the decline in demand

- The cross-county and time-series variation in TV penetration is the main source of identification for the decay of theater profits.
- Monopoly markets are especially useful because strategic interaction is absent there; their exit patterns help pin down the mapping from TV diffusion into demand decline.
- Population growth and post-1949 entry provide additional variation in the decay function.

4.4 How competition and strategic delay are separately identified

- The parameter δ controls how profits fall with the number of active competitors. This shapes how exit hazards vary across monopoly, duopoly, triopoly, and more crowded markets.
- Strategic delay is identified from the timing of exits, not only their total number. For a given demand path, the fraction of exits that occur early versus late is informative about how long theaters wait to learn and outlast rivals.
- The paper’s key intuition is that the overall slope of the exit-rate/competition relationship is mostly about oligopolistic competition, while the clustering and timing of exits reveal the additional war-of-attrition effect.

4.5 How asymmetric information is identified

- The variance of exit values, σ_θ^2 , determines how uncertain theaters are about their rivals’ outside options.
- If this variance were small, theaters would know rivals’ incentives more precisely and would wait less.

- The paper therefore uses early-exit moments and clustering of exits to identify the magnitude of private-information-driven strategic waiting.

4.6 Why the model is computationally credible

- The symmetric equilibrium is unique if it exists, which makes the full-solution estimator operational.
- The game can be solved sequentially because, after each exit, the surviving firms enter a lower-dimensional subgame with a simple continuation value.
- These properties are essential: without them, repeated equilibrium computation inside estimation would be infeasible.

4.7 Relation to the literature

- The paper extends Fudenberg and Tirole's incomplete-information war of attrition from duopoly to oligopoly.
- Relative to the Ericson–Pakes tradition, the environment is nonstationary because demand declines exogenously over time.
- Relative to reduced-form studies of industry decline, the paper offers a structural decomposition of delay into a pure oligopolistic-competition component and a strategic-waiting component.

5 Tables and figures

5.1 Figure 1 and Figure 2: why this industry is a good laboratory

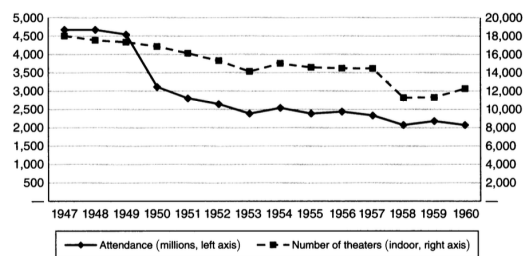


FIGURE 1. MOVIE ATTENDANCE AND NUMBER OF MOVIE THEATERS

Note: Movie theater yearly attendance and the total number of indoor movie theaters.

Source: *The Film Daily Yearbook of Motion Pictures*.

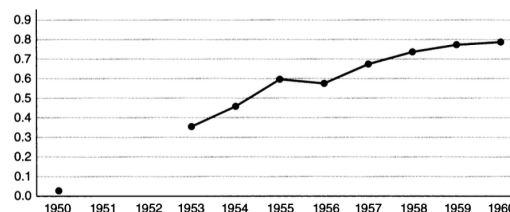


FIGURE 2. AVERAGE OF MARKET-LEVEL TV PENETRATION RATES

Note: These rates are not reported in 1951 and 1952.

Source: Gentzkow and Shapiro (2008a).

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- Figure 1 shows the two aggregate trends motivating the paper: attendance falls sharply and the number of indoor theaters falls more slowly.
- Figure 2 shows the rapid diffusion of TV across local markets. Since this decline is plausibly exogenous to individual theater decisions, it gives the paper a clean driver of market contraction.

- Together, these figures justify the central economic premise: the industry was hit by a persistent, externally driven decline in demand.

5.2 Figure 3 and Table 3: reduced-form evidence for strategic exit

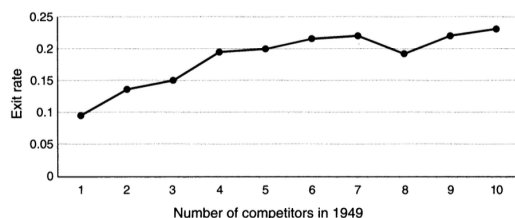


FIGURE 3. EXIT RATE BY NUMBER OF COMPETITORS IN 1949

Sources: *The Film Daily Yearbook of Motion Pictures* and author's calculation.

Parameters	OLS		OLS		IV Regression	
	Estimate	SE	Estimate	SE	Estimate	SE
Constant	0.0845	0.0114	0.0200	0.0163	-0.2272	0.0541
α_m in 1949	—	—	0.0365	0.0071	0.1907	0.0323
$(\alpha_m \text{ in } 1949)^2$	—	—	-0.0025	0.0007	-0.0193	0.0033
Change in TV rate	0.1229	0.0216	0.0813	0.0222	0.0729	0.0262
Change in population	-0.0229	0.0227	-0.0382	0.0226	-0.0369	0.0251
New entrants	0.0706	0.0088	0.0604	0.0089	0.0612	0.0100
Adjusted R^2	0.0350		0.0533		0.0598	

Notes: The dependent variable is the share of theaters in each market that exit during the sample period. For the IV regression, I use demographic variables in Table 2 as instruments for the initial number of competitors and its squared term. The p -value for the F -statistic in the first stage is 0.0000 for both regressions.

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- Figure 3 shows that exit rates rise with the initial number of competitors. This is exactly what a war-of-attrition interpretation predicts in a declining market.
- Table 3 shows that larger increases in TV penetration are associated with higher exit shares, while population growth mitigates exit.
- Adding the initial number of competitors and its square yields a positive coefficient on competitors and a negative coefficient on the square term. The effect remains in IV specifications.
- The reduced-form patterns therefore already suggest that competition structure matters for consolidation.

5.3 Table 1, Table 2, and Table 4: sample composition and market heterogeneity

Number of theaters in 1949	Frequency	Percent
1	451	17.31
2	520	19.95
3	445	17.08
4	369	14.16
5	245	9.40
6	209	8.02
7	143	5.49
8	87	3.34
9	80	3.07
10	57	2.19
Total	2,606	100.00

Variable	Obs.	Mean	SD	Min.	Max.
Population	2,606	22,791	19,627	1,870	194,182
Median age	2,606	4.09	1.03	1.00	7.00
Median family income	2,606	4.91	1.57	0.00	9.00
Urban share	2,606	0.24	0.23	0.00	0.95
Employment share	2,606	0.96	0.02	0.81	1.00
Land area (square miles)	2,606	967	1,289	25	18,573

Note: Median age and median family income are categorical variables.

Sources: Hanes, Michael R., and Inter-university Consortium for Political and Social Research. "Historical, Demographic, Economic, and Social Data: The United States, 1790–2002."

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- Table 1 shows that the sample is dominated by small local markets, which is useful because monopoly and duopoly markets sharply discipline the model.
- Table 2 documents substantial demographic heterogeneity across counties in levels of population, urbanization, income, and land area.
- Table 4 shows wide variation in the initial number of theaters even within fairly similar population bins, motivating the need for unobserved market heterogeneity α_m .

TABLE 4—SUMMARY STATISTICS OF NUMBER OF THEATERS IN 1949 BY POPULATION SIZE

Population of counties	Observations	Mean	SD	Minimum	Maximum
0–10,000	648	2.0	1.1	1	7
10,000–20,000	863	3.0	1.6	1	9
20,000–30,000	473	4.2	1.9	1	10
30,000–40,000	258	5.4	2.2	1	10
40,000–50,000	161	6.2	2.1	1	10
50,000–75,000	129	7.2	1.8	2	10
75,000–100,000	49	8.0	1.7	5	10
100,000–150,000	23	8.3	1.2	6	10
150,000+	2	8.0	1.4	7	9

Sources: Author's calculation based on the population data used in Table 2 and theaters' data from *The Film Daily Yearbook of Motion Pictures*.

5.4 Figure 4 and Figure 5: the equilibrium mechanics

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- Figure 4 illustrates the duopoly policy function. The first exit is delayed because each theater waits for the rival to quit first; the second exit is simply a monopoly stopping problem.
- Figure 5 shows how the policy function is revised once one theater exits in a triopoly. The game restarts with one less competitor and higher continuation profits.
- These figures explain why exit times can cluster and why delays are largest for earlier exits.

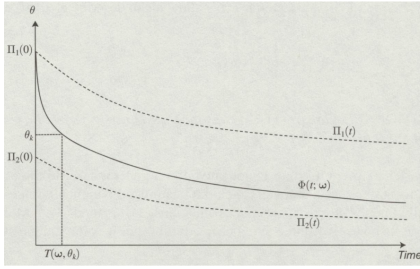


FIGURE 4. POLICY FUNCTION IN THE CASE OF DUOPOLY

Notes: The policy function in the two-player game is given by $\Phi(t; \omega)$. Following this, theater θ_x waits until $T(\omega, \theta_x)$, and in case nobody has dropped out, exits.

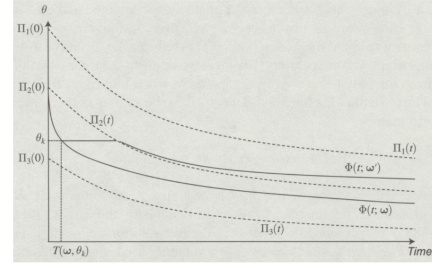


FIGURE 5. POLICY FUNCTION IN THE CASE OF OLIGOPOLY

Notes: The policy function in the three-player game is given by $\Phi(t; \omega)$. After theater θ_x drops out at time $T(\omega, \theta_x)$, the policy function in the two-player subgame is given by $\Phi(t; \omega')$.

5.5 Table 5 and Figure 6: parameter estimates and fit

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- The estimated competition effects are economically meaningful: $\delta = 0.2416$ in the dynamic game and $\delta_e = 0.3121$ in the entry game. Moving from monopoly to duopoly cuts base demand noticeably, and moving to triopoly cuts it further.
- The TV coefficient in the decay function is strongly positive, $\lambda_1 = 0.3715$, confirming that TV diffusion is the main driver of shrinking demand.
- The estimated standard deviation of exit values is large, $\sigma_\theta = 2.413$, implying substantial private-information heterogeneity across theaters.
- Figure 6 shows that the simulated survival profiles line up well with the data across different initial market structures, with only limited mismatches in some larger markets around 1952.

TABLE 5—ESTIMATES OF STRUCTURAL PARAMETERS

Parameters	Estimate	SE
δ (competition in dynamic game)	0.2416	0.0362
δ_e (competition in entry game)	0.3121	0.0167
β_0 (constant)	0.8744	0.0221
β_1 (population)	7.4122	0.2179
β_2 (median age)	1.9634	0.0997
β_3 (median income)	0.8300	0.0186
β_4 (urban share)	-0.3939	0.0118
β_5 (employment share)	-3.4959	0.0282
β_6 (log of land area)	2.4158	0.1060
λ_1 (TV rate)	0.3715	0.0175
λ_2 (change in population)	-0.2220	0.7917
λ_3 (new entrants)	0.5137	0.0945
σ_θ (std. of exit value)	2.4130	0.0901
σ_α (std. of demand shifter)	0.0354	0.0049
$\rho_{\alpha\lambda}$ (corr. coefficient b/w α_m and λ_m)	-0.1892	0.1323

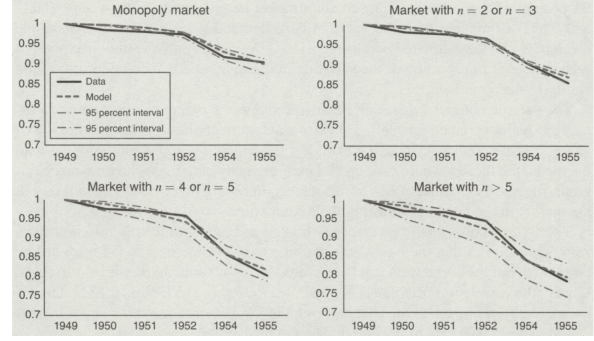


FIGURE 6. MODEL FIT

Notes: I randomly drew a set of structural parameters γ from the estimated asymptotic distribution $N(0, W)$ 200 times, and simulated the survival rate of theaters for each draw. Then, I calculated the 95 percent confidence interval (top 2.5 and bottom 2.5 percent of exit rates) for the exit rate for different market structures.

TABLE 6—DELAY IN EXIT IN YEARS

Market	$t^C - t^R$				$t^* - t^C$			
	Mean	1st	2nd	3rd	Mean	1st	2nd	3rd
$n_m = 2$	2.220	2.220	—	—	0.154	0.154	—	—
$n_m = 3$	2.587	2.967	1.265	—	0.120	0.126	0.102	—
$n_m = 4$	2.483	3.107	1.759	0.729	0.107	0.118	0.093	0.085
All markets	2.577	2.984	2.248	1.498	0.099	0.112	0.084	0.073

Notes: Let t^R , t^C , and t^* be the exit time in the regulator benchmark, in the coordination benchmark, and in a war of attrition equilibrium, respectively. The share of markets in which delays due to oligopolistic competition occur during the sample period is 0.39 for duopoly markets, 0.60 for triopoly markets, and 0.74 for markets with $n_m = 4$. The share of markets in which delays due to strategic behavior occur during the sample period is 0.20 in duopoly markets, 0.31 for triopoly markets, and 0.42 for markets with $n_m = 4$. I calculate the average delay of the first, second, and third exits, as well as the average of all delayed exits by the initial number of competitors.

5.6 Table 6: decomposing delay into oligopoly and strategic behavior

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- The paper compares three exit schedules: the regulator benchmark t^R , the coordinated benchmark t^C , and the equilibrium war-of-attrition outcome t^* .
- Delay due to oligopolistic competition is measured by $t^C - t^R$, while delay due to strategic behavior is measured by $t^* - t^C$.
- Across all markets, the average delay from oligopolistic competition is **2.577 years**, and the additional delay from strategic behavior is **0.099 years**. Their sum is about **2.676 years**, matching the paper's headline "2.7 years."
- In duopoly markets, strategic behavior matters more: the additional strategic delay is 0.154 years, about 6.5% of total delay.

5.7 Table 7 and Table 8: profit costs of strategic interaction

Read:

- Table 7 shows that oligopolistic competition reduces cumulative industry profit by **5.40%** on average relative to the regulator benchmark, while strategic behavior adds another **0.39%** loss on average.
- The costs are larger in more concentrated markets. In duopoly markets, the profit loss due to

TABLE 7—COST OF OLIGOPOLISTIC COMPETITION AND STRATEGIC BEHAVIOR (In percentage difference)

Market	$(Q_m^R - Q_m^C)/Q_m^*$				$(Q_m^C - Q_m^*)/Q_m^*$			
	Mean	5th	Median	95th	Mean	5th	Median	95th
$n_m = 2$	8.21	2.23	7.22	16.42	0.99	0.17	0.70	2.83
$n_m = 3$	5.67	2.57	5.34	10.18	0.37	0.07	0.30	0.91
$n_m = 4$	4.76	2.42	4.56	8.19	0.27	0.04	0.23	0.63
All markets	5.40	2.06	4.68	11.25	0.39	0.05	0.22	1.17

Notes: Let Q_m^R , Q_m^C , and Q_m^* be the total cumulative profit earned by all theaters in market m in the regulator benchmark, in the coordination benchmark, and in a war of attrition equilibrium, respectively. This table shows the summary statistics of these variables by the initial number of competitors.

TABLE 8—COST OF STRATEGIC INTERACTION AND DECLINE IN DEMAND (In percentage difference)

Market	$(Q_m^R - Q_m^C)/Q_m^*$		$(Q_m^C - Q_m^*)/Q_m^*$	
	Slow	Fast	Slow	Fast
$n_m = 2$	7.25	7.21	0.83	0.57
$n_m = 3$	5.29	5.35	0.39	0.23
$n_m = 4$	4.61	4.46	0.27	0.17
All markets	4.44	5.09	0.24	0.20

Notes: Q_m^R , Q_m^C , and Q_m^* are defined in the same way as in Table 7. The table shows the value of the median market for each market structure and speed of decline in demand.

TABLE 9—DELAY IN EXIT IN YEARS WHEN DEMAND IS CONSTANT

Market	$t^* - t^C$			
	Mean	5th	Median	95th
$n_m = 2$	1.841	0.144	1.059	5.342
$n_m = 3$	1.593	0.002	0.908	5.412
$n_m = 4$	1.623	0.002	1.232	4.802
All markets	1.059	0.002	0.303	4.568

Notes: t^C and t^* are defined in the same way as in Table 6. I calculate the model 100 times in each market assuming demand is constant, and average the delay in exit due to strategic behavior by the initial number of competitors. Approximately one-third of the exits occur immediately.

oligopolistic competition is 8.21% and the additional strategic-behavior loss is 0.99%.

- Table 8 shows that the cost of strategic behavior is larger in markets where demand declines slowly. That is intuitive: if future monopoly profits remain high for longer, waiting becomes more attractive.

5.8 Table 9: constant-demand counterfactual

Read:

- When the paper shuts down declining demand and studies a constant-demand environment, strategic delays become much larger.
- The average delay due purely to strategic behavior is about **1.059 years** overall and about **1.841 years** in duopoly markets.
- This counterfactual shows that the relatively small strategic-delay component in the baseline is not because the mechanism is weak; it is because fast demand decline forces theaters to give up sooner.

6 Short summary

- **Conclusion.** The slow exit of movie theaters in the 1950s reflects both ordinary oligopolistic competition and strategic waiting under incomplete information.
- **Identification insight.** Initial market structure and the auxiliary entry model pin down base demand; TV diffusion and monopoly exits identify demand decay; the timing and clustering of exits identify competition and strategic delay.
- **Empirical lesson.** Strategic interactions delay exit by about 2.7 years on average, but most of that delay is due to the noncooperative nature of oligopoly, not to private-information-driven waiting per se.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper develops a tractable oligopoly war-of-attrition model with incomplete information and takes it to data.
- The model fits the theater industry's exit path well and yields a sharp quantitative decomposition of delays in consolidation.
- The headline finding is that strategic interactions matter, but not in the simplistic sense that firms wait forever because of incomplete information. Instead, the main source of delay is the basic externality that each theater does not internalize the benefits its exit creates for rivals.

7.2 Economic intuition

- In a declining industry, someone must exit first. Each theater hopes that another theater will absorb that pain and create more profitable conditions for the survivors.
- If theaters knew each other's outside options perfectly, there would still be delay because firms do not coordinate on the industry-profit-maximizing sequence of exits.
- Private information adds another layer: by surviving, rivals reveal that they may have low exit values, which can induce additional waiting and clustered exits.
- But because demand for movie theaters was falling rapidly due to TV, the scope for purely strategic waiting was limited. This is why most of the delay comes from oligopoly rather than from the information friction.

7.3 Takeaway

This paper is best read as a structural decomposition of slow exit in a declining industry. Its main message is not simply that wars of attrition exist, but that one can estimate them and measure how much they matter. In the theater industry, strategic interaction does delay exit, but the economically dominant force is the noncooperative structure of oligopoly itself. More broadly, the paper shows how a full-solution dynamic game can be estimated in a nonstationary environment and used to translate descriptive industry decline into a disciplined measure of strategic inefficiency.